

Accelerated Mirror Descent for Non-Euclidean Star-convex Functions

Clement Lezane ¹ Sophie Langer ² Wouter M. Koolen ³

¹ANITI Toulouse

²Ruhr University

³University of Twente

ALT 2026

Joint work with



Figure: Sophie Langer and Wouter M. Koolen

Table of Contents

1 Introduction

2 Star-convexity

Table of Contents

1 Introduction

2 Star-convexity

Which acceleration ?

Consider an unconstrained problem

$$\min_{x \in \mathbb{R}^d} f(x)$$

From the literature, we know two kinds of accelerated methods

- Heavy-ball methods

$$x_{t+1} = x_t - \eta \nabla f(x_t) + \gamma(x_t - x_{t-1})$$

- Nesterov's accelerated gradients (optimistic)

$$x_{t+1} = x_t - \eta \nabla f(y_t)$$

What was known in the accelerated framework ?

- Two important conditions on f : f -convex and f having Lipschitz gradients
- Consequences : convergence rate that goes from $\mathcal{O}\left(\frac{1}{T}\right)$ to $\mathcal{O}\left(\frac{1}{T^2}\right)$
- Mirror Descent for non-Euclidean/constrained setting

$$x_{t+1} = \arg \min_x \{ \eta \langle \nabla f(y_t), x \rangle + D_\psi(x, x_t) \}$$

with $D_\psi(a, b) := \psi(a) - \psi(b) - \langle \nabla \psi(b), a - b \rangle$.

Table of Contents

1 Introduction

2 Star-convexity

What is new ?

Definition of τ -star-convex

$$\forall x \in \mathcal{X}, \quad \tau \langle \nabla f(x), x - x_{\star} \rangle \geq f(x) - f(x_{\star})$$

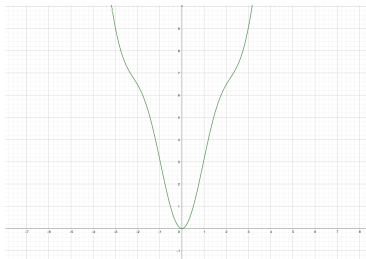


Figure: $f(x) = x^2 + 3\sin^2(x)$

Is the problem much harder ?

Yes, imagine if we were in a finite space with n points, convexity would have $O(n^2)$ constraints

$$\forall x, y \in \mathcal{X}, \quad \langle \nabla f(x), x - y \rangle \geq f(x) - f(y)$$

whereas star-convexity would have only $O(n)$ constraints. And a lot of analysis techniques needs to be reviewed. For instance, the following property would hold for convex functions :

$$\sum_t \alpha_t (f(x_t) - f(x_*)) \leq C \not\Rightarrow f\left(\frac{\sum_t \alpha_t x_t}{\sum_t \alpha_t}\right) - f(x_*) \leq \frac{C}{\sum_t \alpha_t}$$

Novel technique

The binary search technique has been introduced in [Hinder 21]. For simplicity, suppose we have (x_t, y_t) at iteration t , we consider $h_t(\lambda) := f((1 - \lambda)x_t + \lambda y_t) - f(y_t)$ and we aim at finding

$$h_t(\lambda) + \lambda h'_t(\lambda) \leq \epsilon.$$

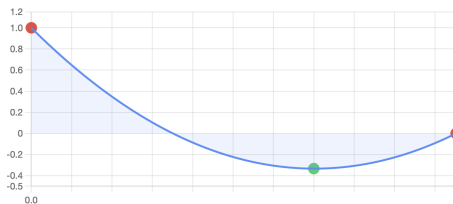


Figure: Binary search on h_t

Which algorithm ?

(1) Solve the PDE and obtain λ_t

$$x_t^{md} = \lambda_t x_t^{ag} + (1 - \lambda_t) x_t$$

(2) Mirror Descent step

$$x_{t+1} = \arg \min_{x \in \mathbb{R}^d} \{ \eta_t \langle \nabla F(x_t^{md}), x \rangle + D_\psi(x, x_t) \}$$

(3) Steepest Descent

$$x_{t+1}^{ag} = \arg \min_{x \in \mathbb{R}^d} \{ \alpha_t \langle \nabla F(x_t^{md}), x \rangle + \frac{\mu}{q} \|x - x_t^{md}\|^q \}$$

Is the convergence rate better ?

Our complexity for (L, κ) weakly smooth problems with regard to p -norm

$$\begin{aligned} 1 < p \leq 2, \quad \mathcal{O}_{\kappa} \left(\left(\frac{L\tau^2 R^{\kappa}}{\epsilon} \right)^{\frac{2}{3\kappa-2}} \log^2 \left(\frac{L\tau R}{\epsilon} \right) \right) \\ p > 2, \quad \mathcal{O}_{p,\kappa} \left(\left(\frac{L\tau^p R^{\kappa}}{\epsilon} \right)^{\frac{p}{\kappa p + \kappa - p}} \log^2 \left(\frac{L\tau R}{\epsilon} \right) \right) \end{aligned}$$

Lower bound from literature for convex problems

$$\begin{aligned} 1 < p \leq 2, \quad \Omega_{\kappa} \left(\left(\frac{LR^{\kappa}}{\epsilon [\ln d]^{\kappa-1}} \right)^{\frac{2}{3\kappa-2}} \right) \\ 2 < p, \quad \Omega_{p,\kappa} \left(\left(\frac{LR^{\kappa}}{\epsilon} \right)^{\frac{p}{\kappa p + \kappa - p}} \right) \end{aligned}$$

Take away message

Three ingredients for general acceleration:

- (1) Momentum effect by solving first order inequality over a line
- (2) (Accelerated) Mirror Descent step
- (3) Steepest Gradient Descent

Future directions :

- Check if momentum effect is satisfied in general setting
- Check if steepest Gradient Descent could be done for a more general metric (an intermedium step to relative smoothness)